FULL LENGTH ARTICLE

Use of spectral decomposition technique for delineation of channels at Solar gas discovery, offshore West Nile Delta, Egypt

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Abstract This paper presents a case study of spectral decomposition of seismic data and how it aids in seismic data interpretation. It is also known as time–frequency analysis where non-stationary signals in time/space are from time/space domain to time/space vs. frequency domain. The frequency domain representation illustrates many important features that are not apparent in time domain representation. Spectral decomposition is a non-unique process for which various techniques exist and newer modified techniques are being discovered. Over the years, spectral decomposition of seismic data has progressed from being a tool for stratigraphic analysis to helping as a Direct Hydrocarbon Indicator (DHI); a potential weapon for reducing dry well drilling. In the coming years, it is expected that spectral decomposition may be able to play a significant role in analyzing time lapse seismic data.

In this paper, spectral decomposition technique is applied to the imaging and mapping of bed thickness, geologic discontinuities and channel delineation at Solar discovery. From the study, two distinctively different channels (gas bearing "Red channel" and water bearing "Yellow channel") were delineated in the area, and are proven by the drilled well, and some stratigraphic features are identified.

1. Introduction

Seismic data, being non-stationary in nature, have varying frequency content in time. Time–frequency decomposition (also called spectral decomposition) of a seismic signal aims to characterize the time-dependent frequency response of subsurface rocks and reservoirs. Spectral decomposition unravels the seismic signal into its constituent frequencies. This allows the interpreter to see amplitude and phase tuned to specific wavelengths, just as a radio can pick out a single station or a prism a single color. Since the stratigraphy resonates at wavelengths dependent on the bedding thickness, the interpreter can image not only subtle thickness variations and discontinuities,

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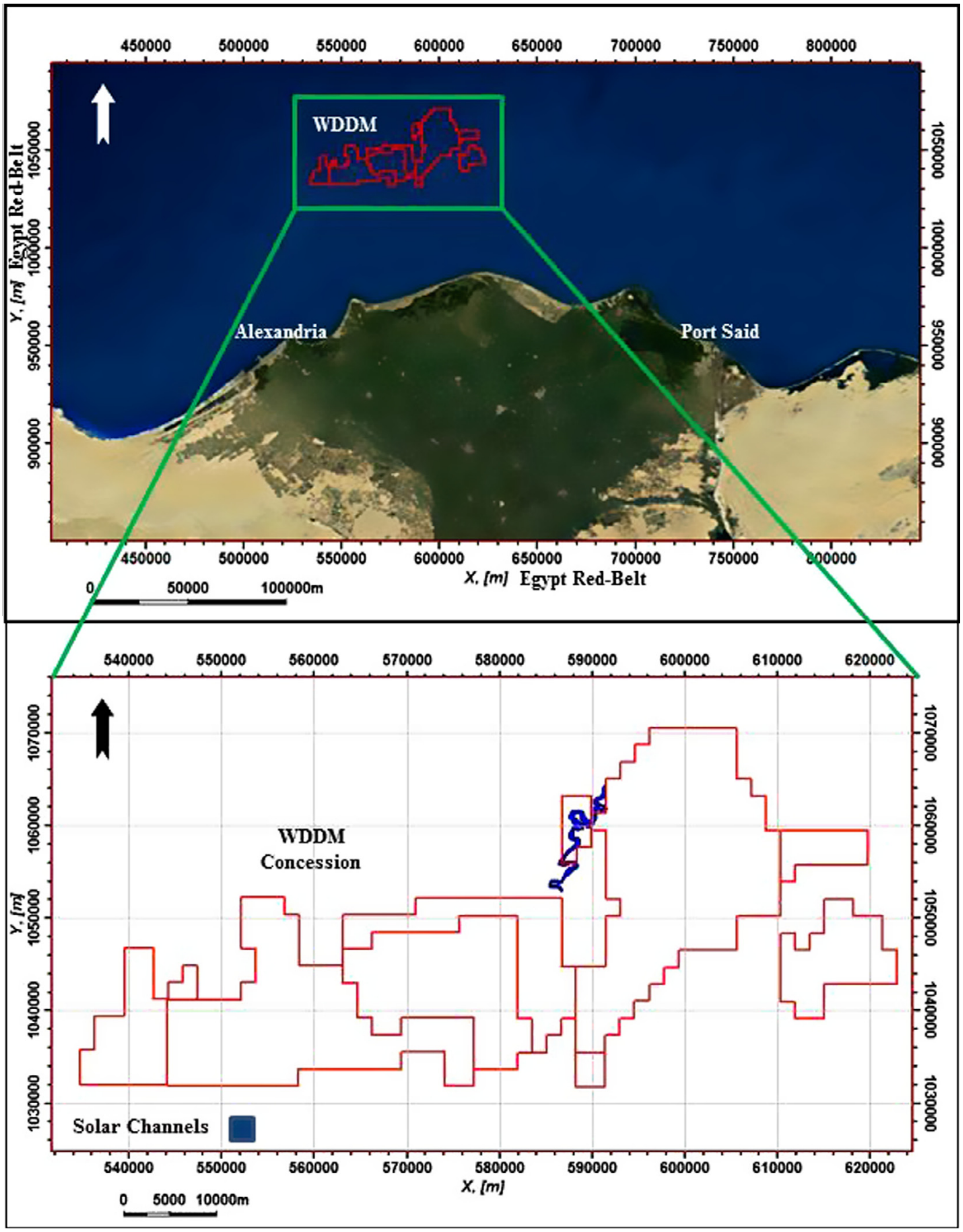

Figure 1 Satellite image showing the location of West Delta Deep Marine (WDDM) concession, offshore Egypt (upper) and an index map showing Solar channel location (lower).

but also accurately predict bedding thickness quantitatively [7]. In addition, since the high-frequency response of a reflector can be attenuated by the presence of compressible fluids, spectral decomposition can also assist in the direct detection of hydrocarbons [1]. These approaches, applied alone to a seismic dataset, can be very enlightening but the results can also be somewhat cryptic. Seismic modeling gives the interpreter a critical insight into tuned attribute maps, and also allows him or her to predicting what a particular bed geometry or thickness trend will look like. Volume visualization and interpretation can enhance the interpretation further still, significantly reducing exploration risk.

2. General geological setting

Egypt is divided into three main petroleum provinces which are Gulf of Suez, Nile Delta and Western Desert. The Nile Delta province is rapidly emerging as a major gas province, highly prolific, with significant Yet To Find (YTF) estimates [3,5,9].

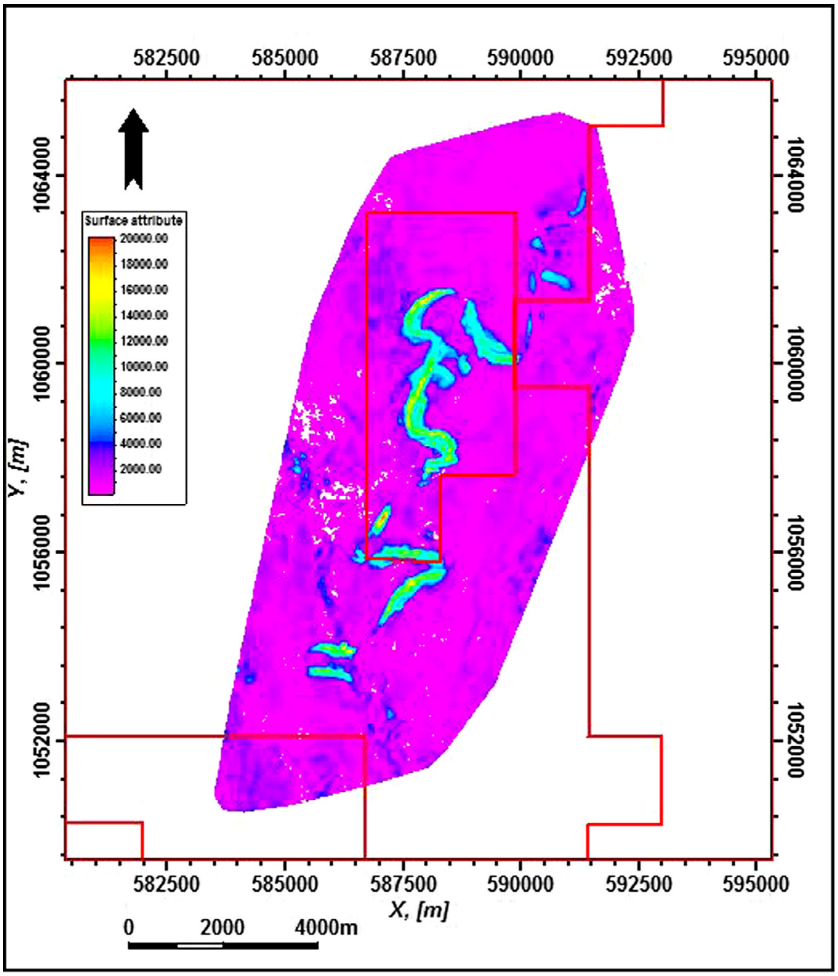

Figure 2 Sum of positive amplitude attribute map (over Solar top Red channel) showing the NE–SW trending Solar channel.

Solar is a Pliocene gas discovery in the West Delta Deep Marine concession (WDDM) located 100 km north off the present day shoreline of Egypt (Fig. 1) in water depths between 800 and 1000 m. WDDM concession is located in the Mediterranean Sea around 110 km NE of Alexandria [4,10]. Two channels have been identified in Solar discovery. In this paper, they are referred to as the Red channel and Yellow channel.

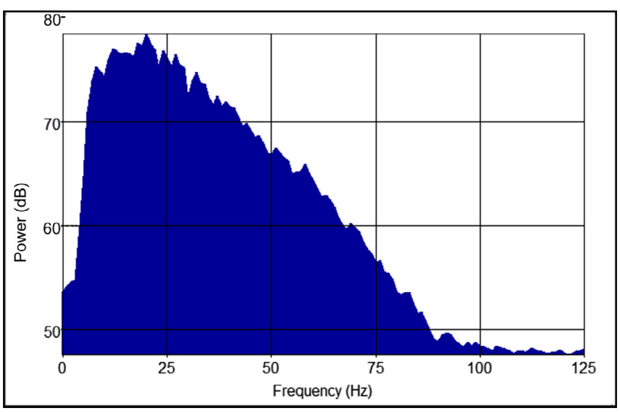

Figure 3 Amplitude spectrum for the full stack seismic data within the interval 2200–3000 ms.

Both channels exhibit a combined stratigraphic/structural trapping style and were deposited in a confined deep marine turbiditic environment (Fig. 2). Laterally both channels are stratigraphically side sealed (pinch out). The overall northward dipping nature of the channel creates dip closure to the north. To the south, stratigraphic closure is interpreted at the head of the channel, this is the highest risk section of the trap. The vertical seal is El-Wastani claystone with some risk of seal capability due to the thinning of the overburden. The well location was optimized to penetrate a relatively high amplitude, thick section at the intersection of the two channels. The Red channel is about 10–12 km long, 400–500 m wide and 40 m average thickness.

3. Conventional data interpretation

The whole field was surveyed during 2006, and its high quality three-dimensional (3D) seismic cube was later processed in 2007. No multiple reflections, or even small multiplicative events, were seen on stacked–migrated seismic sections, particularly in the vicinity of the target reservoir zone.

Data were extensively visualized, colored-contoured and conventionally interpreted using the integrated Petrel reservoir characterization software (Schlumberger Limited, USA). Three time horizons, within which fluid-bearing channels can

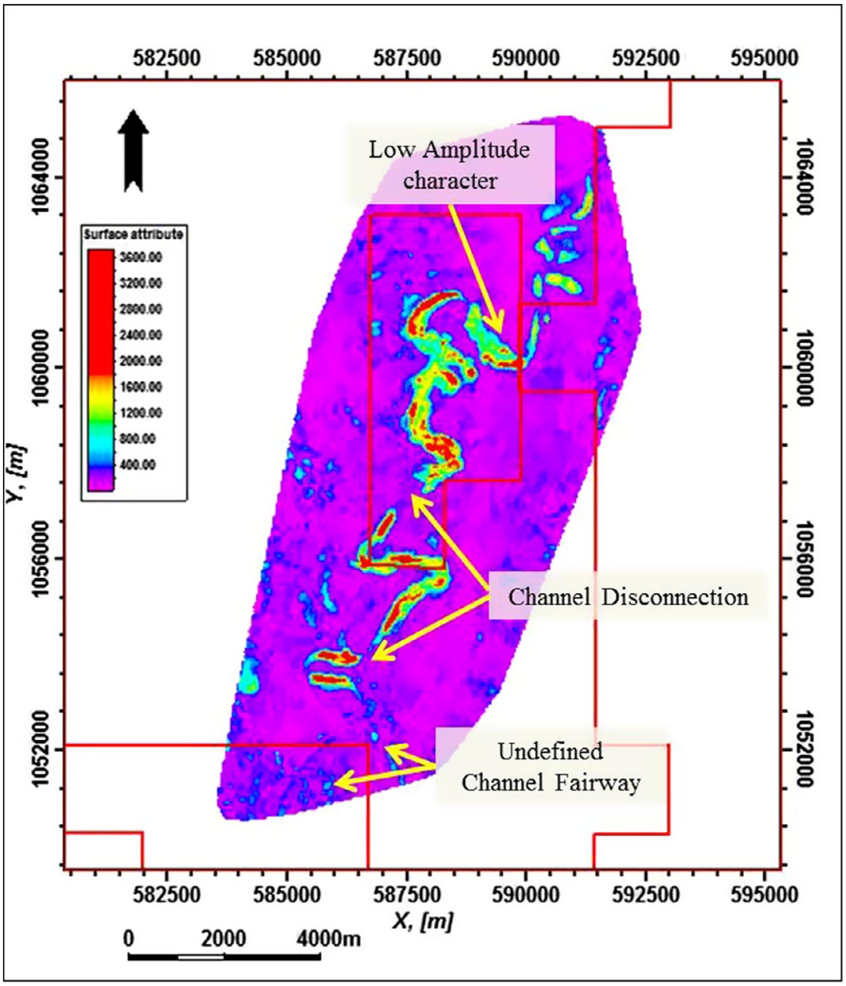

Figure 4 Average magnitude attribute map over the top Solar Red channel shows a low amplitude character in some parts of the channel, discontinuities on the channel fairway, and the poorly defined fairway at the south.

exist, were obviously interpreted in terms of main reflectors; namely, 'Top Red Ch Gas Sand', 'Base Red Ch Gas Sand' and 'Base Red Channel'.

4. Spectral decomposition history

Castagna [1] used matching-pursuit decomposition for instantaneous spectral analysis to detect low-frequency shadows beneath hydrocarbon reservoirs. Peyton [8] used spectral decomposition and coherency to interpret incised valleys. Partyka [7] used windowed spectral analysis to produce discrete-frequency energy cubes for applications in reservoir characterization. Hardy [2] showed that an average frequency attribute produced from sine curve-fitting strongly correlates with shale volume in a particular area. Spectral decomposition transforms the seismic data into the frequency domain via mathematic methods such as Discrete Fourier Transform (DFT), Continuous Wavelet Transform (CWT), and other methods. The transformed results include tuning cubes and a variety of discrete common frequency cubes.

Spectral decomposition has proved to be a robust approach for seismic interpretation. It is used for mapping temporal bed thickness [7], to indicate stratigraphy traps [6], and to delineate hydrocarbon distribution [1].

5. Methodology and results

From the 3D seismic dataset, we know that there are two channels for Solar discovery (Red channel and Yellow channel). The Red channel appears clearly on the seismic and could easily be interpreted. It is sinuous and has a discontinuity in its central part in the form of a dim part of the channel which is attributed to late stage mud-filled channels that mask the seismic amplitudes. The southern part of the channel has two branches and the channel fairway could not be easily defined. The Yellow channel is dimmer than the Red channel and very hard to follow the reflectors for the top and base.

Prior to running the spectral decomposition, the dominant frequency of the seismic data is measured (approximately 25 Hz, Fig. 3). The channel fill tuning frequency may be either greater or less than the overall seismic dominant frequency. Spectral decomposition is run in order to analyze different frequencies of the seismic volume, observes the different bands and examines each band volume.

The spectral decomposition workflow focuses on processing Discrete Fourier Transform (DFT) around a very smooth seismic horizon interpretation, transforming the amplitude or phase data into the frequency domain.

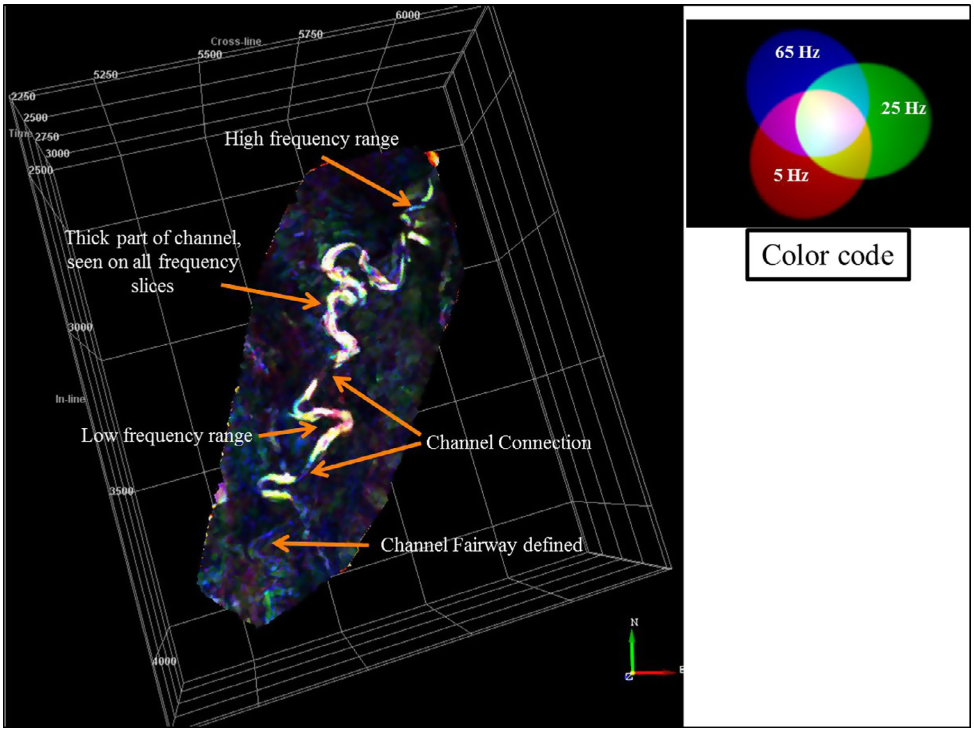

Figure 5 Color blended frequency map over the top Solar Red channel. The map shows the channel connectivity which could not appear on the AAA map, the channel fairway is better defined at the southern part.

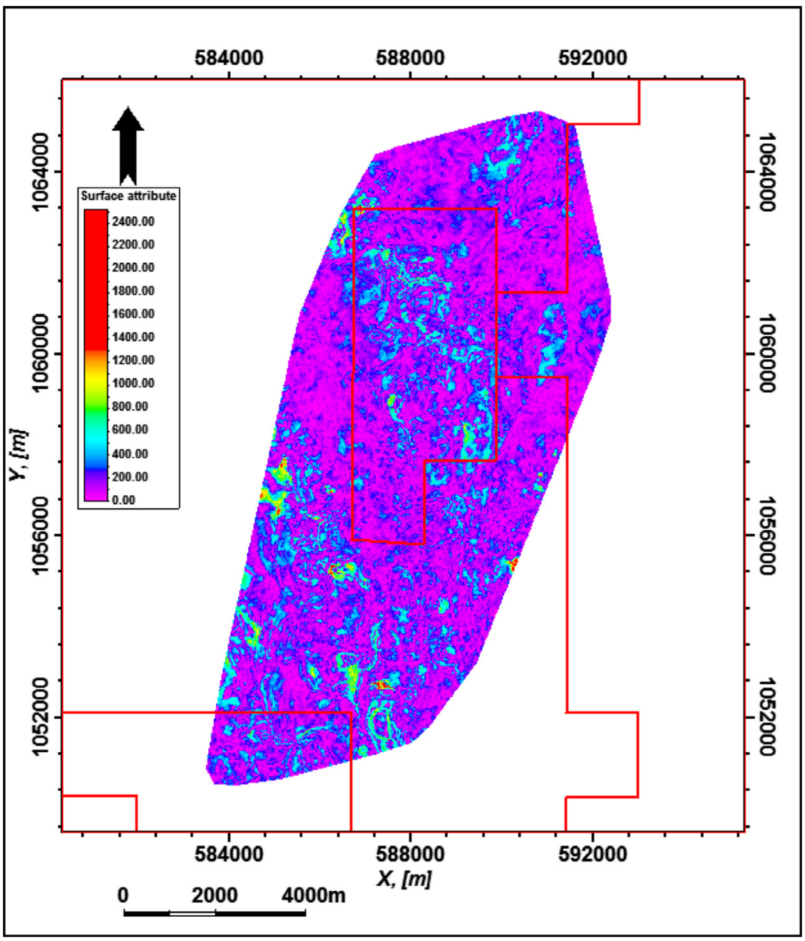

Figure 6 Average Absolute Amplitude (AAA) map along Solar Yellow (water) channel does not describe the reservoir architecture.

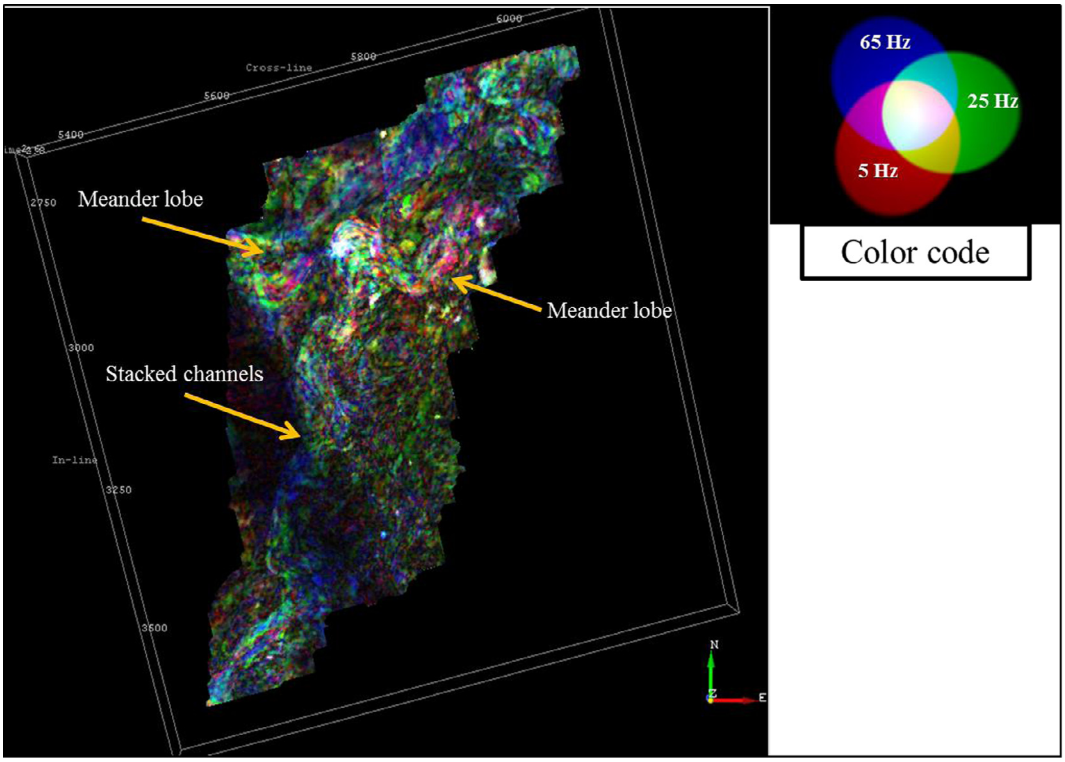

Figure 7 Color blended frequency map along Solar Yellow (water) channel. The map shows the stacked channels which form the Yellow channel.

We used the Red–Green–Blue (RGB) color blended maps; where each color corresponds to a specified frequency range. The three frequency ranges are: 5 Hz (the lowest frequency range in the seismic dataset) in Red color, 25 Hz (the dominant frequency in the seismic dataset) in Green color, and 65 Hz (the highest frequency range in the seismic dataset) in Blue color (Fig. 5).

An average magnitude attribute extraction (Fig. 4), also called Average Absolute Amplitude (AAA), on the top of Solar Red channel shows clear discontinuities. Some parts of the channel exhibit relatively low seismic amplitudes, this could be interpreted as low quality facies or minor faults below seismic resolution that possibly separate the channel into disconnected compartments. The southern part of the channel has no defined fairway.

Comparing the results of the spectral decomposition (Fig. 5) with the AAA map, the dim parts at the middle of the channel on the AAA map appear to coincide with the low frequency range in the spectral decomposition map (in Red color). This means that the channel seems to be connected. The southern part of the channel coincides with the high frequency range on the spectral decomposition map (in Blue color), which defines the channel fairway. Some parts of the channel appear in white color which means they contain all frequency ranges; this corresponds to the thickest parts of the channel.

The same workflow is applied with the aim of investigating features of the lower channel (Yellow channel). AAA map (Fig. 6) and the spectral decomposition map (Fig. 7) are calculated. The spectral decomposition map shows more details compared to the AAA map. The Yellow channel appears to consist of multi-stacked channels, most likely water wet channels.

6. Conclusion

Spectral decomposition helps to identify the channel fairways and delineate the channel continuity at the Red channel (gas bearing sand) while shows water bearing multi stacked channels for the Yellow channel which does not stand out very well in the normal time attribute (AAA map).

The color blending display helps to optimize the well location in terms of targeting the thickest and high quality part of the channel which appears as the white color (a mix of the three RGB frequencies) on the display.

The spectral decomposition provides substantially more detail and fidelity than full bandwidth conventional attributes. It reveals stratigraphic and structural edges as well as relative thickening and thinning. It allows viewing subsurface seismic interference at discrete frequencies.

Acknowledgments

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References

[1] J.P. Castagna, S. Sun, R.W. Seigfried, Lead. Edge 22 (2003) 120–127.

[2] H.H. Hardy, R.A. Beier, J.D. Gaston, Geophysics 68 (2003) 370–380.

[3] C. Harwood, N. Hodgson, M. Ayyad, The application of sequence stratigraphy in the exploration for Plio-Pleistocene hydrocarbons in the Nile Delta, Egyptian General Petroleum Corporation Exploration & Production Conference Proceedings, 1998, pp. 1–13.

[4] H.E. Katamish, N. Steel, A. Smith, P. Thompson, WDDM, The Jewel of the Nile, SPE International, 2005, p. 1, No. 94123.

[5] M.A. Kirschbaum, C.J. Schenk, R.R. Charpentier, T.R. Klett, M.E. Brownfield, J.K. Pitman, T.A. Cook, M.E. Tennyson, Assessment of undiscovered oil and gas resources of the Nile Delta Basin Province, Eastern Mediterranean: U.S. Geological Survey Fact Sheet 2010–3027, 2010, pp. 4.

[6] K.J. Marfurt, R.L. Kirlin, Geophysics 66 (2001) 1274–1283.

[7] G.A. Partyka, J.M. Gridley, J. Lopez, Lead. Edge 18 (1999) 353–360.

[8] L. Peyton, R. Bottjer, G. Partyka, Lead. Edge 17 (1998) 1294–1298.

[9] A. Samuel, B. Kneller, S. Raslan, A. Sharp, C. Pasons, AAPG Bull. 87 (4) (2002) 541–560.

[10] N. Steel, R. Heath, C. Izzat, M. Dodd, A. Samuel, R. Ramadan, M. Nashaat, West Delta Deep Marine, Offshore Nile Delta – Aggressive Exploration Program Commercialized Through ELNG, PESGB, 2003.